LAB 7: Fuzzing for Input Validation Bugs

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Repository:

github.com/chaturyaprabhakar/L7_63_house_painting_quote (LAB7 folder)

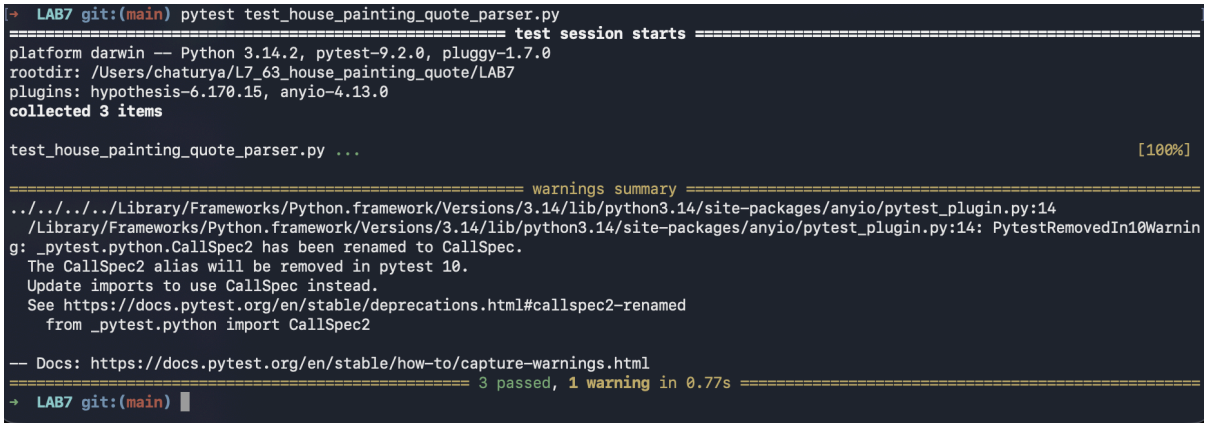
SS1 — Buggy module run (python3 house_painting_quote_parser.py)

Crash triggered: entering a room size with no 'x' causes a ValueError in room_area().



SS2 — Hypothesis tests run against the buggy module (failing)

pytest test_house_painting_quote_parser.py — all three no-crash tests fail, surfacing AttributeError, ValueError and ExceptionGroups across clean_colour_name, room_area and total_wall_area.



```
→  LAB7 git:(main) ✗ pytest test_house_painting_quote_parser.py
========================================= test session starts ==========================================
platform darwin -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/chaturya/L7_63_house_painting_quote/LAB7
plugins: hypothesis-6.170.15, anyio-4.13.0
collected 3 items

test_house_painting_quote_parser.py FFF                                                        [100%]

================================================ FAILURES ================================================
_____________________________________________ test_clean_colour_name_no_crash _____________________________________________

s = None

    @given(st.text() | st.none())
    def test_clean_colour_name_no_crash(s):
        try:
>           clean_colour_name(s)

test_house_painting_quote_parser.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

name = None

    def clean_colour_name(name):
        """
        Turns a colour name into a lower-case code, e.g. 'Ocean Blue' -> 'ocean_blue'.
        Assumes name is a non-empty string.
        """
>       return name.strip().lower().replace(' ', '_')
               ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

house_painting_quote_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_colour_name_no_crash(s):
                ^^^

test_house_painting_quote_parser.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_clean_colour_name_no_crash(s):
        try:
            clean_colour_name(s)
        except Exception as e:
>           assert False, f"clean_colour_name crashed with: {e}"
E           AssertionError: clean_colour_name crashed with: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_clean_colour_name_no_crash(
E               s=None,
E           )
E           Explanation:
```

```
test_house_painting_quote_parser.py:9: AssertionError
_____________________________________________ test_room_area_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 12, in test_room_area_no_crash
  |     def test_room_area_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 14, in test_room_area_no_crash
    |     room_area(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 14, in room_area
    |     return int(length) * int(width)
    |                ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 16, in test_room_area_no_crash
    |     assert False, f"room_area crashed with: {e}"
    | AssertionError: room_area crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_room_area_no_crash(
    |     s='x',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 14, in test_room_area_no_crash
    |     room_area(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 13, in room_area
    |     length, width = text.lower().split('x')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 16, in test_room_area_no_crash
    |     assert False, f"room_area crashed with: {e}"
    | AssertionError: room_area crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_room_area_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 14, in test_room_area_no_crash
    |     room_area(s)
    |     ~~~~~~~~~^^^
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 13, in room_area
    |     length, width = text.lower().split('x')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'lower'
    |
    | During handling of the above exception, another exception occurred:
    |
```

```
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 21, in test_total_wall_area_no_crash
    |     total_wall_area(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 21, in total_wall_area
    |     return sum(int(area) for area in text.split(';'))
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 21, in <genexpr>
    |     return sum(int(area) for area in text.split(';'))
    |                ~~~^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 23, in test_total_wall_area_no_crash
    |     assert False, f"total_wall_area crashed with: {e}"
    | AssertionError: total_wall_area crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Falling test case: test_total_wall_area_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 21, in test_total_wall_area_no_crash
    |     total_wall_area(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/house_painting_quote_parser.py", line 21, in total_wall_area
    |     return sum(int(area) for area in text.split(';'))
    |                                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/chaturya/L7_63_house_painting_quote/LAB7/test_house_painting_quote_parser.py", line 23, in test_total_wall_area_no_crash
    |     assert False, f"total_wall_area crashed with: {e}"
    | AssertionError: total_wall_area crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Falling test case: test_total_wall_area_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=============================================== warnings summary ===============================================
../../../../Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/anyio/pytest_plugin.py:14
  /Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/anyio/pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================================== short test summary info ============================================
FAILED test_house_painting_quote_parser.py::test_clean_colour_name_no_crash - AssertionError: clean_colour_name crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_house_painting_quote_parser.py::test_room_area_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_house_painting_quote_parser.py::test_total_wall_area_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================================= 3 failed, 1 warning in 0.35s =============================================
➜  LAB7 git:(main) ×
```

SS3 — Fixed module run (python3 house_painting_quote_parser.py)

Same malformed inputs now handled gracefully — functions return '' or 0 instead of crashing.

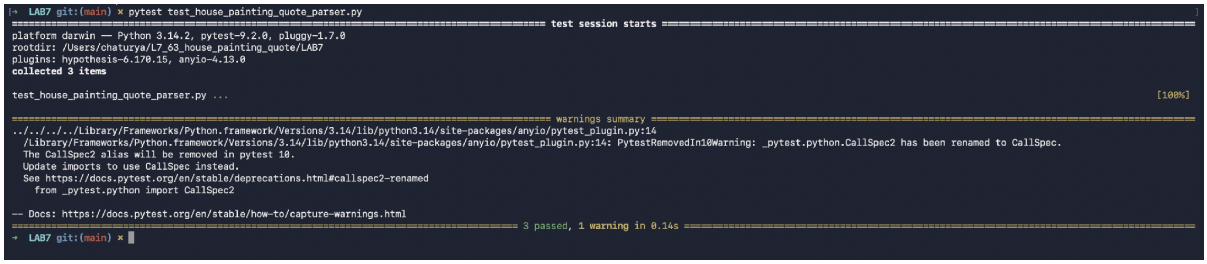
```
➜  LAB7 git:(main) × python3 house_painting_quote_parser.py
==== Running clean_colour_name ====
Enter a paint colour name (e.g. Ocean Blue):
Colour code:

==== Running room_area ====
Enter room size as LENGTHxWIDTH in feet (e.g. 12x10):
Room area (sq ft): 0

==== Running total_wall_area ====
Enter wall areas separated by ; (e.g. 120;95;110):
Total wall area (sq ft): 0
➜  LAB7 git:(main) ×
```

SS4 — Hypothesis tests run against the fixed module (passing)

pytest test_house_painting_quote_parser.py — 3 passed, 0 failed.



Bugs Found and Fixes Applied

clean_colour_name(name)

Bug: crashed with AttributeError on non-string input (e.g. None) because .strip() assumes a string.

Fix: checks isinstance(name, str) first and returns '' for non-string or empty input instead of crashing.

room_area(text)

Bug: crashed with ValueError('not enough values to unpack') when there was no 'x' in the input, and with AttributeError on None input.

Fix: validates the input is a string, confirms exactly one 'x' is present, confirms both parts are digit strings, and returns 0 for anything that doesn't fit this shape instead of unpacking blindly.

total_wall_area(text)

Bug: crashed with ValueError on empty elements between semicolons (e.g. '120;;100') and AttributeError on None input.

Fix: validates the input is a string, strips and skips empty or non-numeric segments, and sums only valid whole numbers.